

ZS-A32

The Planck-Pivot Extremal Ladder: Four Corners, One Modular Depth

Friedmann-Forced Squares of the Register-Clock Depth $e^{(-2\pi Q)}$, the Telomere Rational $n_{\text{Hubble}} = 4QA = 1540/437$, the Executed Pre-Registered Anti-Numerology Monte Carlo ($p = 0.50\%$), and the Honest Residuals (H-CLK) and C_{UV}

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Date: July 2026

Theme / Code: Astrophysics / ZS-A32 · Executes the pre-registration first frozen in ZS-F38 v1.0 Appendix C and reconciled in ZS-F38 v1.1 (MC3 deferral), and the ZS-M47 §12.3 cosmology handover · Downstream of ZS-F38 / ZS-A31 / ZS-M46 / ZS-M47 / ZS-F37 / ZS-A30

Version: 1.1

Verification: 34/34 exact/numerical PASS + 7/7 consistency guards (zs_a32_verify_v1_1.py, SHA-256 eb142f7f...; sympy exact + mpmath 50-digit + 10^6 -draw Monte Carlo) | **MC EXECUTED: $p = 0.50\% \leq 5\%$** (pre-registered single target; robustness audits: dedup-invariant, simplest-core 3.5%, alternative nulls $\leq 1.0\%$) | Zero new fitted Z-Spin parameters (one disclosed external Λ CDM background package: $\bar{M}_P, \rho_\Lambda, H_0, t_U, \Omega_\Lambda$ — Friedmann-related, not independent) | **(A, Q, dim Z) = (35/437, 11, 2) LOCKED.**

§0. Abstract

ZS-A31 §6.3 pre-registered a single frontier hypothesis for the cosmological hierarchy — the metric/Kähler scale sits at modular depth Q below the reduced Planck scale, $M_K/\bar{M}_P = e^{(-2\pi Q)}$, the 2π forced by Borchers–Wiesbrock, with the observed effective scale $M_{\text{eff}} = C_{\text{UV}}^{(1/4)} M_K$ — and ZS-F38 v1.0 Appendix C froze the anti-numerology design (an 88-formula universe, the tolerance band $\frac{1}{4} \ln 4 \equiv$ the C_{UV} prior band $[1/4, 4]$, the single target $(c, q) = (2, 11)$), as reconciled in ZS-F38 v1.1, while deferring execution to this paper. We execute it, and publish the pre-registration certificate — target freeze (A31 v1.2) → universe freeze (F38 App. C) → execution (this script, hash and seed of record) — as Appendix C. Against the null range $[0, \ln(\bar{M}_P/H_0)] = [0, 138.68]$ — the widest hierarchy realized in nature, built from observed quantities only — the pre-registered single-target chance rate is $p = 0.50\% \leq 5\%$ (analytic and 10^6 -draw Monte Carlo agree; the observed $|\Delta \ln| = 0.054$ is sharper still, at 0.08%), while the look-elsewhere coverage of the full 88-formula universe is 28.8% — the quantitative demonstration that pre-registration, not proximity, carries the verdict. The verdict is audit-stable: deduplicating the universe leaves the look-elsewhere coverage exactly unchanged (interval-union invariance), the simplest-core stratum of the formula grammar covers only 3.5%, and the primary p stays $\leq 1.0\%$ under triangular, Beta(2,2), and log-uniform alternative nulls — all published as audits, none as verdict basis. Per the corpus promotion rule, corner (W) is promoted HYPOTHESIS-weak → HYPOTHESIS-strong; it is simultaneously DERIVED-CONDITIONAL under the ZS-F38 registry.

The structural content is the extremal ladder around the Planck pivot: the stage extrema (the largest length and time) and the actor extremum (the faintest wave scale) are one number. Theorem A32.1 (Friedmann-Forced Squares) shows that, given (W), the Friedmann equation and the corpus factor $\rho_\Lambda = \frac{1}{2}(1260/4807)\omega^2 M_{\text{eff}}^4$ force $\ln(\bar{M}_P/\rho_\Lambda) = 8\pi Q + O(1)$ and $\ln(\bar{M}_P/H_0) = \ln(R_H/\ell_P) =$

$4\pi Q + 0.562 - \frac{1}{2} \ln C_{UV}$, with $\ln(t_U/\bar{t}_P)$ the same shifted by $\ln(H_0 t_U)$: the vacuum, length, and time corners are arithmetic consequences of the single wave-corner hypothesis — there is no multiple testing to correct, which is this consolidation’s strongest anti-numerology defense. The observed corners land at 276.64 ($8\pi Q$; 0.07%), 138.68 and 138.63 ($4\pi Q$; 0.33%, 0.29%), and the C_{UV} read off the wave and Hubble corners agrees to 0.06% — a closure identity, not an additional datum, since the input is one Λ CDM background package (H_0 , Ω_Λ , ρ_Λ are Friedmann-related), not three independent observations: $C_{UV} \approx 1.24$, inside the declared band, awaiting the full-1PI computation (ZS-F36, OPEN; the target specification is fixed in §7). Precisely stated: the wave, vacuum, length, and time hierarchy corners are controlled by one registered modular depth; the particle floor is separated into the determinant family and is not claimed as part of the same π -family (§6).

Theorem A32.2 dissolves the ZS-A31 gate F-A31.2 puzzle: on the ZS-M3/F10 telomere ladder $t_n = \bar{t}_P e^{(n\pi/A)}$, the Hubble corner sits at the exact rational rung $n = 4QA = 1540/437 = 3.52403$ — by the arithmetic identity $4\pi Q = (4QA)(\pi/A)$ — dressed by the computed Friedmann offset to $n_{obs} = 3.53556$ (identity to 3×10^{-5}); the ladder was never supposed to hit an integer, and A31’s quoted 3.57 is reproduced exactly as the unreduced-convention reading. The particle corner is honestly separated: the corpus-derived electroweak exponent is $(38/9)(11 \ln 2 + \ln 3) = 36.831$ on the determinant ($\ln 2 / \ln 3$ torsion) lattice, not the modular π -family; the Reidemeister value $35\pi/3 = 36.652$ sits 0.49% away and is registered as a non-identified proximity (NON-CLAIM). The model-level C_{int} of ZS-F37 App. D is executed on the ZS-F38.T2 Hardy chain with $\varepsilon_{C_{int}} = 0$ exactly; the physical C_{int} stays OPEN behind $(H-CLK)/(H-\Sigma 2)$. Net status of the ZS-A22 barrier B3: an OPEN two-gate debt $\{(H-CLK) \text{ discharge}, C_{UV} \text{ 1PI}\}$, its statistical leg now closed. An Export Theorem (Appendix D) states the corner arithmetic corpus-independently. Verification: 34/34 PASS + 7/7 guards. Zero new fitted Z-Spin parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED.

Epistemic Status Legend

Tag	Meaning
PROVEN	Explicit proof or exact machine verification; no undischarged assumption.
PROVEN (model)	Exact in the named model realization; physical identification not implied.
IMPORTED-PROVEN	Proven in the external literature and used without re-proof; cited.
DERIVED	Follows from PROVEN results by stated steps; no new parameter.
DERIVED-CONDITIONAL	Derived modulo explicitly named, falsifiable conditions (here: (W), the ZS-F38 registry, disclosed Λ CDM inputs).
HYPOTHESIS-strong	Pre-registered numerical relation that has passed its pre-registered anti-numerology Monte Carlo ($p \leq 5\%$); mechanism pending.
HYPOTHESIS-weak	Pre-registered, falsifiable relation; anti-numerology verification pending.
REGRESSION	Fixed from observation behind the derivation $\perp$ regression firewall (the one dimensional input, inherited from ZS-A31).
CONSISTENCY-CHECK	Arithmetic or numerical agreement that does not by itself certify a theorem.
OBSERVATION	Empirical or packaging regularity; structural origin pending; no claim made.
INHERITED-OPEN	An upstream OPEN item registered and re-scoped, not addressed here.
NON-CLAIM	Explicitly outside this paper’s scope.
OPEN	A genuine remaining gap, honestly registered.

§1. Introduction: the two handovers

This paper is the cosmology consumer of the register-clock program. It discharges two inherited mandates and does nothing else beyond them.

§1.1 The ZS-F38 handover (executed here)

ZS-F38 v1.0 delivered the register-clock consolidation: T1 (clock identity $\Rightarrow$ the democratic register measure $\rho = I_Q/Q$ by unique ergodicity, conditional on (H-CLK) and (H-cycle)), T2 (the Q-unit inclusion chain in the Hardy model, with the Jones–Longo even subchain $\Gamma = U(2)$), and T3 (the depth selection $t^* = Q$, DERIVED-CONDITIONAL on the registry {(H-CLK), (H-cycle), KH1–KH4, parent injectivity, NDC3}). Its Appendix C froze the anti-numerology Monte Carlo design — universe, band, single target, decision rule — and its guard MC3 explicitly deferred execution to ZS-A32 to protect pre-registration integrity. ZS-F38 v1.1 subsequently reconciled the registry wording, decomposed (H-eval), and made the p-statistic hierarchy explicit (primary p_{single} , secondary formula-count/look-elsewhere) while preserving the frozen design and this paper’s executed values unchanged; §4 executes that design without modification, and cites the design as “first registered in ZS-F38 v1.0 Appendix C, reconciled in ZS-F38 v1.1” throughout.

§1.2 The ZS-M47 §12.3 handover (registered, re-scoped)

The ZS-M47 §12.3 handover table assigned four rows to ZS-A32. They are operator-algebraic constructions; the four-corner arithmetic of this paper consumes none of them. Per the one-paper-one-ledger discipline they are registered INHERITED-OPEN and re-scoped, with their tests intact:

Table 1. ZS-M47 §12.3 items assigned to ZS-A32 — disposition.

Item	Deliverable	Disposition here
(H-2D) designation	conjugate seam (λ^* , equal modular speed) + tensor Borchers triple	INHERITED-OPEN $\rightarrow$ ZS-M48/ZS-F39 (KH-transfer test intact)
P2 envelope; ($\star_w$); (TT)	candidate constructions E1/E2/E3; the weight-level program	INHERITED-OPEN $\rightarrow$ ZS-M48/ZS-F39
Lorentzian bulk spectral theory	registered vdD–P–R toolset; the AdS ₃ wedge net	INHERITED-OPEN $\rightarrow$ ZS-M48/ZS-F39
$t^* = Q$ depth selector	canonical tunnel, no rung selected (NON-CLAIM in M47)	superseded by ZS-F38.T3 (DERIVED-CONDITIONAL); consumed in §2

Locked inputs: $A = 35/437$, $Q = 11$, $(\dim Z, \dim X, \dim Y) = (2, 3, 6)$; $\omega = \theta = \arg \lambda^* = 2.2592495540$ (ZS-M1); the reduced Planck convention $\bar{M}_P = M_P/\sqrt{(8\pi)} = 2.435 \times 10^{27}$ eV (ZS-A31 Locked Inputs). Disclosed external inputs (standard, cited): $\rho_\Lambda^{1/4} = 2.24$ meV, $H_0 = 67.36$ km s⁻¹ Mpc⁻¹, $t_U = 13.797$ Gyr (Planck 2018 Λ CDM); $\Omega_\Lambda = 83/121$ is consumed from ZS-A30 (DERIVED-CONDITIONAL; the Planck value 0.6889 shifts the Friedmann offset by < 0.003 and is carried as a cross-check).

§2. The four-corner extremal ladder

§2.1 Definition D1 (the Planck pivot as a two-sided extremum)

The corpus’ stage/actor separation (The Book, terminology contract: the Z-sector is the stage; the Z-Spin is the action on it) induces a two-sided extremal reading of the Planck scale. *Stage quantities* — length and time — have their global minima at the pivot ($\ell_P, \bar{t}_P$): the external anchors are the Doplicher–Fredenhagen–Roberts minimal-length model (gravitational stability forbids sub-Planckian localization) and the corpus’ own ZS-F30 Lorentzian seam scaffolding. *Actor quantities* — wave and particle, the Z-Spin’s two sector readings — have their global maxima at the pivot: the Margolus–Levitin/Lloyd quantum speed limits cap the clock rate and mass of any actor at the Planck

scale. The four opposite extremes are then cosmological, and the claim of this paper is that all four are priced by one modular depth:

- (W) wave minimum: $M_K/\bar{M}_P = e^{(-2\pi Q)}$ [the single hypothesis; ZS-A31 (8.3), pre-registered]
(V) vacuum density: $\ln(\bar{M}_P^4/\rho_\Lambda) = 8\pi Q + O(1)$
(L) length maximum: $\ln(R_H/\ell_P) = 4\pi Q + O(1)$
(T) time maximum: $\ln(t_U/\bar{t}_P) = 4\pi Q + O(1)$

with (V), (L), (T) carrying no hypothesis content of their own. The particle floor is treated separately and honestly in §6: its corpus-derived exponent lives on a different lattice.

§2.2 Theorem A32.1 (Friedmann-Forced Squares) [DERIVED-CONDITIONAL]

Statement. Assume (W): $M_{\text{eff}} = C_{UV}^{1/4} \bar{M}_P e^{(-2\pi Q)}$ with $C_{UV} \in [1/4, 4]$ (the ZS-F36 full-1PI factor, OPEN; the band is the declared prior). Given the corpus vacuum factor $\rho_\Lambda = \frac{1}{2}(1260/4807)\omega^2 M_{\text{eff}}^4 = 0.668952 M_{\text{eff}}^4$ (ZS-F33/F36/M44/A31 chain, DERIVED-CONDITIONAL on register-trace normalization) and the Friedmann equation $H_0^2 = \rho_\Lambda/(3\Omega_\Lambda \bar{M}_P^2)$ (IMPORTED, standard), the three remaining corners follow by arithmetic:

$$\begin{aligned} \ln(\bar{M}_P^4/\rho_\Lambda) &= 8\pi Q - \ln C_{UV} - \ln 0.668952 = 8\pi Q - \ln C_{UV} + 0.402, \\ \ln(\bar{M}_P/H_0) &= 4\pi Q - \frac{1}{2} \ln C_{UV} + \frac{1}{2} \ln(3\Omega_\Lambda/0.668952) = 4\pi Q - \frac{1}{2} \ln C_{UV} + 0.562, \\ \ln(t_U/\bar{t}_P) &= \ln(\bar{M}_P/H_0) + \ln(H_0 t_U), \quad \ln(R_H/\ell_P) = \ln(\bar{M}_P/H_0) \quad (\text{exact unit identity}). \end{aligned}$$

Proof. Substitute (W) into the vacuum factor: $\rho_\Lambda = 0.668952 C_{UV} \bar{M}_P^4 e^{(-8\pi Q)}$, giving the first line. Insert into Friedmann: $H_0 = \bar{M}_P e^{(-4\pi Q)} C_{UV}^{1/2} \sqrt{0.668952/(3\Omega_\Lambda)}$, giving the second (with $\Omega_\Lambda = 83/121$, the offset is $\frac{1}{2} \ln(3 \cdot (83/121)/0.668952) = 0.5619$). The third line is the definition of the age factor $H_0 t_U$ and the identity $R_H = c/H_0$, $\ell_P = c \bar{t}_P$. Every step is a substitution; no new number enters. $\square$

The anti-numerology structure of the theorem is its point: the exponent multipliers $\{1, 2, 4\}$ on $2\pi Q$ are not three matches but one — the quartic root in $M_{\text{eff}} = C_{UV}^{1/4} \bar{M}_P \leftrightarrow$ the Friedmann square $\leftrightarrow$ the quartic density. A single pre-registered hypothesis prices all four corners; there is no trial to correct for. (Guard PR3.) The same discipline applies to the inputs: the input package is one Λ CDM background package, not three independent observations — once H_0 , Ω_Λ , and $\bar{M}_P$ are chosen, ρ_Λ is Friedmann-related — therefore the C_{UV} cross-corner agreement of §2.3 is a closure identity, not an additional datum, and this paper never counts it as one.

§2.3 The corner table

Table 2. The Planck-pivot extremal ladder (reduced convention; all deviations against the raw modular exponent, before the computed Friedmann/ C_{UV} dressing).

Corner	Quantity	Predicted exponent	Observed	Dev.	Status
(W) wave min	$\ln(\bar{M}_P/M_{\text{eff}})$, $M_{\text{eff}} = 2.476 \text{ meV}$	$2\pi Q = 69.115$	69.061	0.079%	HYP-strong (§4)
(V) vacuum	$\ln(\bar{M}_P^4/\rho_\Lambda)$; $8\pi Q = 120.07 \text{ dex}$	$8\pi Q = 276.460$	276.644	0.067%	CONSISTENCY
(L) length max	$\ln(R_H/\ell_P) = \ln(\bar{M}_P/H_0)$	$4\pi Q = 138.230$	138.683	0.327%	CONSISTENCY
(T) time max	$\ln(t_U/\bar{t}_P)$; $H_0 t_U = 0.9505$	$4\pi Q = 138.230$	138.632	0.291%	CONSISTENCY

The dressed residuals close on one number. From (W): $C_{UV} = (M_{eff}/(\bar{M}_P e^{(-2\pi Q)}))^4 = 1.244$. From (L) via Theorem A32.1: $C_{UV} = e^{\{2(4\pi Q + 0.562 - \ln(\bar{M}_P/H_0))\}} = 1.244$. Their agreement to 0.06% is an arithmetic identity (both descend from the same ρ_Λ observation through Ω_Λ), not an independent test — but it demonstrates that after the single hypothesis and the computed offsets, *zero residual freedom remains*. The famous “ 10^{120} ” of the cosmological-constant problem is, in this reading, the identity $8\pi Q/\ln 10 = 120.07$ dex in the reduced convention (the quoted 10^{122} – 10^{123} arises from the unreduced M_P^4 , a pure convention spread of $4 \ln \sqrt{8\pi} = 6.4$ in $\ln$). (Checks CN1–CN5, FR1–FR4, LD3.)

§3. Theorem A32.2: the telomere rational $n_{\text{Hubble}} = 4QA$

ZS-M3 §8 / ZS-F10 derived the timescale hierarchy $t_n = \bar{t}_P e^{(n\pi/A)}$ — the telomere ladder — and ZS-A31 §4 registered as gate F-A31.2 that the Hubble time lands at the non-integer $n_{\text{Hubble}} \approx 3.57$, so “the hierarchy does not pin H_0 .” This section shows the puzzle was mis-posed: the modular depth ladder and the telomere ladder are commensurate at a rational, not an integer, and the rational is exact.

Statement. (i) [PROVEN, arithmetic] $4\pi Q = (4QA) \cdot (\pi/A)$, with $4QA = 1540/437 = 3.5240274\dots$ exactly. (ii) [DERIVED-CONDITIONAL on (W) + Theorem A32.1] The observed telomere coordinate of the Hubble corner is the rational rung dressed by the computed Friedmann offset:

$$n_{\text{obs}} \equiv \ln(\bar{M}_P/H_0)/(\pi/A) = 4QA + [0.562 - \frac{1}{2} \ln C_{UV}]/(\pi/A) = 3.52403 + 0.01154 = 3.535571 ,$$

against the measured 3.535565 — an identity to 3×10^{-5} (check TL4). The integer expectation is rejected by 0.46 rungs, forty times the entire C_{UV} band width in n -units (0.0088); the rational lands inside it. (iii) [VERIFIED] ZS-A31’s quoted 3.57 is the unreduced-convention reading: $(\ln(\bar{M}_P/H_0) + \frac{1}{2} \ln 8\pi)/(\pi/A) = 3.5767$ (check TL3). The convention was fixed reduced in A31’s own Locked Inputs; the residual 3.57 in its §4 prose traces to the M3/F10-era unreduced t_P and is resolved here of record.

Reading. Gate F-A31.2 is not falsified but re-posed: the telomere ladder’s rung variable was never forced to be an integer by any corpus theorem — integer rungs were an aesthetic expectation. What the register clock supplies is the depth lattice $\pi \cdot \mathbb{Q}(A, Q)$: exponents of the form $c\pi q$ and $n\pi/A$ with c, q, n built from the locked constants. The Hubble corner’s coordinate $1540/437$ — numerator $4 \cdot Q \cdot 35 = 1540$, denominator 437 — is as structural as an integer, and it is the *same single hypothesis* as corner (W), re-expressed (no new registration; guard PR3).

§4. The executed pre-registered Monte Carlo

§4.1 The frozen design and the declaration order

The design is the one first registered in ZS-F38 v1.0 Appendix C and reconciled in ZS-F38 v1.1, unmodified (its verification items MC1–MC3 machine-froze it): universe U of $|U| = 88$ formulas — $\{e^{(-c\pi q)}: c \in \{\frac{1}{2}, 1, 2, 4\}, q \in \{2, 3, 6, 11, 22, 28, 33, 35, 49, 121\}\} (40) \cup \{A^k\}_{k=1..40} (40) \cup \{e^{(-n\pi/A)}\}_{n=1..8} (8)$; tolerance band $|\Delta \ln| \leq \frac{1}{4} \ln 4 = 0.3466$, which is exactly the $C_{UV} \in [1/4, 4]$ prior band; single pre-registered target $(c, q) = (2, 11)$, i.e. $e^{(-2\pi Q)}$, inherited from ZS-A31 v1.2; decision rule $p \leq 5\%$. Two declarations were made in this paper’s script *before any p was evaluated* (guard MC0 — named DECLARATIVE-GUARD of record, because an ordering declaration inside a script is a statement, not a machine-verifiable fact; the machine-checkable

chronology evidence — version strings, script hash, seed, dates — is the Pre-registration Certificate, Appendix C): the primary statistic is the single-target chance rate p_{single} (the target predates the universe: A31 v1.2 registered it before F38 built U; the universe exists to quantify the look-elsewhere that *would* apply absent pre-registration); and the null range is L uniform on $[0, \ln(\bar{M}_P/H_0)] = [0, 138.68]$ — the widest hierarchy realized in nature, constructed from observed quantities only (any sub-Planckian physical scale in our universe lies between H_0 and $\bar{M}_P$).

§4.2 Results

Table 3. Monte Carlo execution (10^6 draws; analytic values exact).

Statistic	Value	Rule	Reading
p_{single} (PRIMARY, pre-registered target)	0.500% (analytic); 0.499% (MC)	$\leq 5\%$: PASS	chance rate for the one registered formula
p at observed $ \Delta \ln = 0.0545$	0.079%	reported	sharper, not adjudicated (band was the registered tolerance)
p_{universe} (look-elsewhere, all 88)	28.76% exact; 28.68% (MC)	reported	matching some formula is unremarkable — pre-registration carries the verdict
sensitivity: range $[0, 160] / [30, 110]$	0.43% / 0.87%	$\leq 5\%$: PASS	decision stable under alternative declared ranges
audit: deduplicated universe ($ U $ 88 $\rightarrow$ 82)	$p_{\text{universe}} = 28.76\%$ (unchanged)	audit	interval-union coverage is dedup-invariant — degeneracy retention neither flatters nor harms
audit: simplest-core stratum ($c \in \{1,2\}, q \in \{2,3,6,11\}$)	coverage = 3.50%; target $\in$ core	audit	under any complexity weighting favouring simple formulas, look-elsewhere collapses below 5%
audit: alternative nulls (triangular / Beta(2,2) / log-uniform)	1.00% / 0.75% / 0.20%	audit	primary decision stable under null-model choice

Verdict. The pre-registered target is hit at $|\Delta \ln| = 0.0545$ — well inside the band — with a single-trial chance rate of 0.50%. Per the corpus promotion rule (convergent exploration + no corpus collision + pre-registered MC executed at $p \leq 5\%$), corner (W) is registered **HYPOTHESIS-strong**. The 28.8% universe coverage is published alongside it deliberately: it is the measured size of the numerological temptation this program's pre-registration discipline exists to defeat — had the target been chosen after seeing the data, no promotion would be permitted (the ZS-M47 CY4 precedent, $p = 0.352$, applied its own rule against identification; this paper applies the same rule for it). The three audit rows are published on an *audit basis, not a verdict basis*: the verdict rests solely on the pre-registered primary; the audits quantify how the secondary statistics and the null model behave under alternative design choices an external critic could demand — the target $2\pi Q$ is checked to be literally a member of the rebuilt π -family (MC1, non-tautological containment), and the degenerate pair $\pi \cdot 22 = 2\pi \cdot 11$ is confirmed retained at the target value as registered (MC7).

Dual status. The promotion is the statistical leg. Independently, corner (W) is DERIVED-CONDITIONAL under the ZS-F38 registry $\{(H\text{-CLK}), (H\text{-cycle}), KH1\text{--}KH4, \text{parent injectivity, NDC3}\}$ plus the C_{UV} band: if the registry discharges and the full 1PI returns $C_{UV} \approx 1.24$, the corner closes to DERIVED-CONDITIONAL with no band; if the registry falls, the HYPOTHESIS-strong statistical reading survives it (and vice versa: an observational exit from the band kills both). The two legs fail independently — the honest architecture.

§5. The model-level C_{int} execution (ZS-F37 App. D on the ZS-F38.T2 chain)

ZS-F37 App. D specified the finite computation that closes the clock-gate matching C_{int} in its ZS-M46 T7b form: match the iteration step of the seam to the modular step of the inclusion and report

the residual norm ε_{C_int} . The physical execution requires the actual ZS-F31 GKLS / ZS-A24 core data and remains scoped to the Foundations successor. What ZS-F38.T2 newly supplies is the canonical model in which the protocol can be executed exactly: the Hardy-model Q-unit inclusion chain, where the seam is the unit Abel translation and the chain step is multiplication by z .

Executed result [PROVEN (model)]. On the F38.T2 chain the two steps are the same map: the branch-consistent Abel increment of the corpus seam is $u(fz) - u(z) = 1$ at twelve independently-computed Koenigs orbit points to $< 10^{-12}$ (check CI1, re-verifying M46 GEO2-class data), and the chain shift is $z \cdot (z^k) = (z^{k+1})$ for all $k \leq Q$ exactly in symplectic (check CI2). The matching residual is therefore $\varepsilon_{C_int}^{(model)} = 0$ identically — not small, zero, because in the model the clock identity holds by construction. What this does and does not close: it converts C_int from “an unformulated computation” to “computed-zero in the canonical model”; the physical C_int — whether the actual Z-Spin seam channel realizes this model — is exactly the content of (H-CLK)/(H-Σ2) and remains OPEN (guard CI3). The model execution sharpens the target: any future physical computation returning $\varepsilon_{C_int} \neq 0$ would falsify the clock identity, not merely miscalibrate it.

Discharge criterion (fixed here for ZS-F39/ZS-M48). Compute $\varepsilon_{C_int}^{(phys)}$ on the actual ZS-F31 GKLS / ZS-A24 seam-channel core data. If it is zero within the registered norm, (H-CLK) is discharged and the F38 registry keystone falls into place; if not, this paper’s statistical leg survives but the F38 mechanism leg fails — the two-leg architecture of §4.2 already prices both outcomes. A32 deliberately does not attempt this computation: pushing (H-CLK) inside a consumer paper would weaken both.

§6. The particle corner and the two exponent families

The actor’s particle floor is real but lives on a different lattice, and this paper declines to force it onto the modular one. The corpus-derived electroweak hierarchy is the Ray–Singer factorized determinant (ZS-F24 v2.0, Table A1): $\ln(\bar{M}_P/v) = (38/9)(11 \ln 2 + \ln 3) = 36.8314$, giving $v = 245.93$ GeV (the ZS-S4 locked value; check PC1), and the gauge-sector mass gap is the Q-singlet projection $m_{gap} = v \cdot A/Q = 1.791$ GeV (ZS-S7, DERIVED-CONDITIONAL; lattice 0^{++} glueball 1.73 ± 0.05 GeV, $+1.2\sigma$; check PC3). The exponent family is the determinant (torsion) lattice $\{a \ln 2 + b \ln 3 : \text{rational weights}\}$ — not $\pi \cdot \mathbb{Q}(A, Q)$. The Reidemeister value $35\pi/3 = 36.6519$ sits 0.49% from the derived exponent (check PC2), and guard PC4 confirms it is not a member of the pre-registered π -family universe (minimum distance 0.87 in $\ln$): the proximity is registered and explicitly *not identified* — the §6-level instance of the ZS-M44 §5.4 trap lesson. Unifying the two families is a NON-CLAIM of this paper (attempted and terminated in exploration; Appendix A, dropped candidate 6).

Table 4. The two exponent families (honest separation).

Family	Members (exponent, value)	Provenance
Modular π -family (this paper)	$2\pi Q = 69.115$ (W); $4\pi Q = 138.230$ (L, T); $8\pi Q = 276.460$ (V); rational rung $4QA = 1540/437$	Borchers–Wiesbrock $2\pi \times$ register depth Q (ZS-F38.T3); multipliers $\{1, 2, 4\}$ Friedmann-forced
Determinant family (cited)	$(38/9)(11 \ln 2 + \ln 3) = 36.831$ (EW); $+\ln(Q/A)$ to the gap	Ray–Singer torsion on the register complex (ZS-F24); Q-singlet projection (ZS-S7)

Packaging observation [OBSERVATION / NON-CLAIM]. The de Sitter entropy of the observed horizon is $\ln S_{dS} = 2 \ln(\bar{M}_P/H_0) + \ln 8\pi^2 = 281.73 = 8\pi Q + 5.27$ (check OB1) — the additive constants are the computed $8\pi^2$ and Friedmann dressing — and Lloyd’s computational budget of the

universe ($\sim 10^{120}$ operations) is $\ln \approx 276 \approx 8\pi Q$. In the register-clock reading these are the information-capacity face of the same vacuum corner, not independent evidence; no claim is registered.

§7. Status ledger and the B3 update

Table 5. Claim ledger (v1.0).

Claim	Status
Corner (W): $M_K/M_P = e^{(-2\pi Q)}$ ($M_{\text{eff}} = C_{UV}^{1/4} M_K$)	HYPOTHESIS-strong (MC executed, $p = 0.50\%$) // DERIVED-CONDITIONAL (ZS-F38 registry + C_{UV} band)
Theorem A32.1 (Friedmann-Forced Squares)	DERIVED-CONDITIONAL (on (W); Friedmann IMPORTED; Λ CDM inputs disclosed)
Corners (V), (L), (T): exponents $8\pi Q$, $4\pi Q$, $4\pi Q$	CONSISTENCY-CHECK (no independent hypothesis; deviations 0.07–0.33% before dressing)
C_{UV} cross-corner agreement (1.244 vs 1.244, 0.06%)	CONSISTENCY-CHECK (arithmetic identity, not a test)
$C_{UV} \approx 1.24$ itself	INFERENCE only; full 1PI OPEN (ZS-F36); Q-absorption forbidden (A31 discipline)
Theorem A32.2(i): $4\pi Q = (4QA)(\pi/A)$, $4QA = 1540/437$	PROVEN (exact arithmetic)
Theorem A32.2(ii): $n_{\text{obs}} = 4QA + \text{dressing}$ (3×10^{-5})	DERIVED-CONDITIONAL (same single hypothesis re-expressed)
A31's $n \approx 3.57 = \text{unreduced-convention reading}$	VERIFIED (convention resolved of record)
$\epsilon_{C_{\text{int}}} = 0$ on the F38.T2 Hardy chain	PROVEN (model); physical C_{int} OPEN behind (H-CLK)/(H- Σ_2)
M47 §12.3 rows: (H-2D); ($\star$ w)/(TT); Lorentzian bulk	INHERITED-OPEN $\rightarrow$ ZS-M48 / ZS-F39
Particle-corner family unification	NON-CLAIM (35 π /3 proximity registered, not identified)
S dS / Lloyd-budget packaging	OBSERVATION / NON-CLAIM
$M_{\text{eff}} = 2.476 \text{ meV}$ (the one dimensional input)	REGRESSION (inherited from ZS-A31, firewall intact)

The B3 update. ZS-A22 barrier B3 — no action-level mechanism fixes H_0/M_P — entered ZS-A31 as a void and leaves this paper as a *two-gate debt*: (i) discharge (H-CLK) at action level (the register-clock identity; ZS-F38's single keystone), and (ii) compute the full-1PI C_{UV} (ZS-F36; the inferred 1.24 is the target it must hit). Its statistical leg — could the depth-Q proximity be chance? — is closed at $p = 0.50\%$. Nothing here violates the ZS-A17/A27/A28 no-go: no scale is derived; the single dimensional anchor remains the one A31 regression; every new statement is a ratio.

C_{UV} Target Specification [registered here; the second gate's contract]. The full-1PI computation (ZS-F36 line) must return $C_{UV} = 1.244 \pm \delta$, in the same reduced-Planck convention, without absorbing any residual into Q (the A31 discipline), where δ is the propagated uncertainty of the Λ CDM package (the Hubble-corner reading shifts by $|\Delta \ln C_{UV}| = 2 \cdot \ln(73/67.36) = 0.161$ at the SH0ES extreme, i.e. $C_{UV} \in [1.24, 1.46]$ across the tension — still a sharp target inside the $[1/4, 4]$ band). This fixes the successor's goal with zero freedom: A32 is not the paper that closes C_{UV} ; it is the paper that leaves C_{UV} nowhere to hide.

§8. Falsification gates

F-A32.1 (mathematical, immediate). The exact identities fail: $4\pi Q \neq (4QA)(\pi/A)$, or the Theorem A32.1 substitution algebra breaks, or 29/29 does not reproduce. Immediate rejection.

F-A32.2 (statistical integrity). Any demonstration that the MC design (universe, band, target, rule) differs from the ZS-F38 App. C freeze, or that the primary statistic was chosen after evaluation, voids the §4 verdict and demotes (W) to HYPOTHESIS-weak.

F-A32.3 (observational). Future ρ_{Λ}/H_0 determinations moving the wave corner outside $[2\pi Q - \frac{1}{4} \ln 4, 2\pi Q + \frac{1}{4} \ln 4]$ (equivalently C_{UV} outside $[1/4, 4]$) falsify (W) in both legs. The current

Hubble-tension spread ($\text{SH0ES } H_0 \approx 73$) shifts $\ln(\bar{M}_P/H_0)$ by -0.081 and n_{obs} by -0.002 — inside all bands; a resolution of the tension does not threaten the corner, only sharpens C_{UV} .

F-A32.4 (registry collapse). An action-level refutation of (H-CLK), (H-cycle), or KH1–KH4 kills the DERIVED-CONDITIONAL leg; the HYPOTHESIS-strong statistical leg survives independently and would then demand a different mechanism.

F-A32.5 (the C_{UV} gate). A computed full-1PI C_{UV} outside $[1/4, 4]$, or inside the band but incompatible with the inferred 1.24 beyond its propagated uncertainty, falsifies the exact-depth reading. A computed $C_{\text{UV}} \approx 1.24$ closes the corner to DERIVED-CONDITIONAL without band — the promotion path.

F-A32.6 (telomere). Any future corpus theorem forcing integer telomere rungs re-opens F-A31.2 against this paper; the rational reading stands or falls with (W).

F-A32.7 (family separation). Any post-hoc identification of $35\pi/3$ with the electroweak exponent absent a derivation is void by this paper’s own registration (PC2/PC4).

F-A32.8 (upstream). Re-opening of the M44 register-trace chain or ZS-A30’s $\Omega_\Lambda = 83/121$ recomputes the Friedmann offset; the theorem survives on the Planck Ω_Λ (shift < 0.003) but the affected checks must be re-run of record.

§9. Non-claims

This paper does not derive any dimensionful scale (the ZS-A27/A28 Buckingham- π no-go is inviolate; the one dimensional input remains the A31 regression). It does not compute C_{UV} (inferred only; absorbing the 0.08% wave-corner gap into Q is forbidden by the A31 pre-registration discipline). It does not discharge (H-CLK) or any ZS-F38 registry condition. It makes no claim on the dark-energy equation of state $w(z)$ (ZS-A22/A28 scope: $w = -1$ constant; the DESI phantom-divide reading is A22’s). It does not unify the determinant and modular exponent families, does not claim the particle floor is “the” global actor minimum (the gap statement is gauge-sector-scoped), and registers the $S_{\text{dS/Lloyd}}$ numerology-adjacent packaging as OBSERVATION only.

§10. Verification

zs_a32_verify_v1_1.py (SHA-256 eb142f7fef7cab78bef68f3129454dc97e89892f9362f91054e388ca8ba0bff8): **34/34 exact/numerical PASS + 7/7 guards** (sympy exact rational/symbolic; mpmath 50 digits; numpy 10^6 -draw MC, seed 11; analytic p-values seed-independent; < 30 s). Blocks: LD1–LD4 (locked echoes; the exact commensurability identity), CN1–CN5 (four-corner numerics), FR1–FR4 (Friedmann squares; C_{UV} cross-corner identity), TL1–TL4 (telomere rational, convention resolution, dressing identity), PC1–PC3 + guard PC4 (particle corner, family separation), MC0 [DECLARATIVE-GUARD]–MC7 (the executed pre-registration: MC1 non-tautological target membership, MC7 degeneracy retention, primary $p_{\text{single}} = 0.500\%/0.499\%$, secondary $p_{\text{universe}} = 28.76\%/28.68\%$, sharper 0.079%, sensitivity, verdict), RB1–RB3 (v1.1 robustness audits: dedup invariance exact, simplest-core 3.50%, alternative nulls 1.00%/0.75%/0.20%), ET1 (Export Theorem symbolic, App. D), CERT (runtime self-hash + chronology DECLARATIVE-GUARD, App. C), OB1 (packaging), CI1–CI2 + guard CI3 (model $C_{\text{int}} = 0$), PR1–PR3 (zero-parameter, non-reversal, single-trial guards). The suite does not machine-certify imported theorems (Borchers–Wiesbrock, Friedmann, DFR, Margolus–Levitin/Lloyd) — cited, not re-proven.

§11. Conclusion

One hypothesis, registered before its universe was built, prices the four extremes of the observed universe: the faintest wave at $e^{(-2\pi Q)}$ below the Planck pivot, the vacuum density at its Friedmann-forced fourth power, the largest length and the longest time at its square — and the telomere ladder's stubbornly non-integer Hubble rung resolves into the exact rational $4QA = 1540/437$ the moment the two ladders are read as one lattice $\pi \cdot \mathbb{Q}(\mathbf{A}, \mathbf{Q})$. The pre-registered Monte Carlo returns $p = 0.50\%$ for the one registered target while measuring, at 28.8%, exactly how large the numerological temptation would have been without the registration — the discipline and its justification in a single table. The through-line extends by one clause: the Z-sector supplies the stage; the Z-Spin supplies every action on it; the stage's two-dimensionality counts the legs; *and the register clock's depth $\mathbf{Q}$, struck once per seam step at the Borchers rate 2π , prices every extreme the universe attains*. What remains is exactly what the ledger says: one keystone identity (H-CLK) at action level, and one finite IPI number that already knows its target. $(\mathbf{A}, \mathbf{Q}, \dim \mathbf{Z}) = (35/437, 11, 2)$ LOCKED.

Acknowledgements & Code Availability

The verification suite `zs_a32_verify_v1_1.py` reproduces all $34 + 7$ checks including the executed pre-registered Monte Carlo and the v1.1 robustness audits (fixed seed; analytic values independent of the seed; the script reports its own SHA-256 at runtime for the Appendix C certificate). This work used AI tools (Anthropic Claude) for corpus and external-literature search, cross-paper integration, symbolic/numerical verification, and drafting, under Kenny Kang's editorial direction. The author assumes full responsibility for all content.

Appendix A — Deep-exploration record (Steps 0–5)

Step 0 (long list, 7). (1) The Friedmann-squares theorem as direct consumption of ZS-F38.T3 (the corner table); (2) the telomere rational $4QA$ reinterpretation; (3) the pre-registered MC execution under the frozen design; (4) the physical C_{int} execution on the full ZS-F31 GKLS / ZS-A24 core data; (5) the model-level C_{int} execution on the F38.T2 Hardy chain; (6) a particle-corner unification attempt ($35\pi/3 \leftrightarrow \pi$ -family bridge); (7) holographic/information packaging (S_{dS} , Lloyd budget).

Dropped. (4) — requires operator data outside this astrophysics consumer's scope; routed to the Foundations successor (ZS-F39/M48) with the F37 App. D protocol intact. (6) — attempted and terminated honestly: the corpus-derived EW exponent is 36.8314 on the determinant lattice (F24 Table A1 rows 10–11, by construction of $v = 245.93$ GeV); $35\pi/3 = 36.6519$ sits 0.49% away with no derivation connecting them; identification refused, NON-CLAIM registered (§6; the null is retained as PC2/PC4, the M47 route-5 pattern). (7) demoted to a one-paragraph OBSERVATION (§6).

Step 1 (MECE issue list, 4, by impact). I1 = Friedmann squares + corner table {1}; I2 = MC execution {3}; I3 = telomere rational {2}; I4 = model C_{int} + packaging {5, 7}.

Step 2 (issue tree). I1 $\rightarrow$ (I2, I3) — both consume the corner data; I4 parallel, consuming only F38.T2.

Step 3 (statuses). I1a corner (W): HYPOTHESIS-weak (inherited) $\rightarrow$ HYPOTHESIS-strong post-MC // DERIVED-CONDITIONAL under the F38 registry. I1b Theorem A32.1: DERIVED-CONDITIONAL. I1c corners (V)/(L)/(T): CONSISTENCY-CHECK. I1d C_{UV}

cross-corner: CONSISTENCY-CHECK (identity). I2 MC: EXECUTED, primary $p = 0.50\% \leq 5\%$. I3a 4QA arithmetic: PROVEN. I3b dressing identity (3×10^{-5}): DERIVED-CONDITIONAL. I3c A31 3.57 convention: VERIFIED. I4a $\epsilon_{C_int}(\text{model}) = 0$: PROVEN (model). I4b physical C_int : OPEN. I4c packaging: OBSERVATION/NON-CLAIM.

Step 4 (convergence). Cycle 1 assigned 11 node statuses. Cycle 2 changed 2: (i) the telomere deviation 0.33% was found to be the *computed* Friedmann dressing, not noise — TL4 was rewritten from a band check into the 3×10^{-5} identity (a strengthening); (ii) the particle corner was moved from the π -family table to the determinant family after the F24 Table A1 audit showed the derived exponent is 36.8314, not $35\pi/3$ (an honesty correction; it created guard PC4). Cycle 3 changed 0. Sequence 11 $\rightarrow 2 \rightarrow 0$: converged; MECE retained.

Step 5 (scoring). Converged + corpus-non-colliding (A17/A27/A28 no-go, A31 firewall, A22/A28 w-scope, F38 registry all intact) + pre-registered anti-numerology MC *executed* at $p = 0.50\% \leq 5\% \Rightarrow$ the promotion rule's strongest available grade for a numerical corner: HYPOTHESIS-strong. Against ZS-A31 (the predecessor consolidation): the B3 debt is reduced from a void to two named gates, the F-A31.2 puzzle is dissolved rather than deferred, the 3.57/3.54 convention discrepancy is resolved of record, and the paper adds zero numerical hypotheses beyond the one it inherited — the whole increment is structure and statistics, which is what a consolidation should be.

v1.1 integration cycle (external review of v1.0). A second deep-exploration pass integrated the v1.0 external review. Long list (7): F38 v1.1 registry reconciliation wording; Pre-registration Certificate (App. C); MC verification hardening (non-tautological MC1 membership, MC7 degeneracy, MC0 renamed DECLARATIVE-GUARD); robustness audits (dedup / complexity-stratified / alternative nulls); the one- Λ CDM-package non-independence statement; the C_{UV} Target Specification; the Export Theorem (App. D). Dropped: attempting the physical C_int or the IPI C_{UV} inside A32 (both would inflate scope and weaken the consumer architecture; routed to ZS-F39/M48 and the ZS-F36 line with their contracts fixed here). MECE (4): I1 pre-registration evidence (certificate + hardening) $>$ I2 robustness audits $>$ I3 scope/wording conservatism (v1.1 citation, Λ CDM package, abstract precision) $>$ I4 forward-fixing contracts (C_{UV} target, discharge criterion, Export Theorem). Convergence: cycle 1 assigned 9 node statuses; cycle 2 changed 1 (the complexity-weighting audit was re-designed from an arbitrary token-weight scheme to the parameter-free simplest-core stratum — an honesty improvement: no weight function to tune); cycle 3 changed 0. Sequence 9 $\rightarrow 1 \rightarrow 0$: converged. Verdict basis unchanged throughout: every v1.1 addition is evidence packaging, audit, or contract — zero new hypotheses, zero changed numbers.

Appendix B — Monte Carlo execution detail

Universe (88, frozen): the π -family $\{e^{(-c\pi q)}\}$ over $c \in \{\frac{1}{2}, 1, 2, 4\}$, $q \in \{2, 3, 6, 11, 22, 28, 33, 35, 49, 121\}$ (the corpus constant set: sector dimensions, Q and its doublings, $|V_Y - F_Y| = 28, 3Q$, the A-numerator 35, $7^2, Q^2$); the A-powers $\{A^k\}$, $k = 1..40$ ($\ln A = -2.5243$); the telomere family $\{e^{(-n\pi/A)}\}$, $n = 1..8$ ($\pi/A = 39.2250$). Degenerate values (e.g. $\frac{1}{2} \cdot 22 = 1 \cdot 11$) are retained as registered — deduplication would shrink coverage and flatter the verdict.

Null and statistics. $L \sim \text{Uniform}[0, 138.683]$; the width is $\ln(\bar{M}_P/H_0)$, observed quantities only. Primary: $p_{\text{single}} = 2 \cdot (\frac{1}{4} \ln 4) / 138.683 = 0.500\%$ analytic; 0.499% at 10^6 draws. Secondary: $p_{\text{universe}} = \text{exact interval-union coverage of all in-range bands} = 28.76\%$ (10^5 -draw MC: 28.68%).

Sharper: $2 \cdot 0.0545 / 138.683 = 0.079\%$. Sensitivity: widths 160 and 80 give 0.43% and 0.87%. Decision rule $p \leq 5\%$: PASS on the primary and on every sensitivity variant.

Why the single-target statistic is primary. The target $e^{(-2\pi Q)}$ was pre-registered in ZS-A31 v1.2, before the F38 universe existed; the universe was constructed afterward precisely to measure the look-elsewhere a non-registered search would incur. Adjudicating on p_{universe} would penalize the hypothesis for a search that never happened; publishing p_{universe} alongside prevents the converse abuse. Both numbers are of record.

Robustness audits (v1.1; audit basis). (i) Deduplication: the 88-formula universe contains 6 duplicated exponent values (e.g. $\pi \cdot 22 = 2\pi \cdot 11$ at the target itself); removing them ($|U| \rightarrow 82$) leaves p_{universe} exactly unchanged, because an interval union does not double-count — retention is provably neutral (RB1). (ii) Complexity stratification: instead of an arbitrary token-weight scheme, the parameter-free simplest-core stratum $\{c \in \{1, 2\}, q \in \{2, 3, 6, 11\}\}$ (unit and Borchers-double multipliers over the four locked primaries; 7 distinct values; the target is a member) covers only 3.50% of the null range — under any complexity weighting favouring simple formulas the look-elsewhere concern collapses below the 5% threshold (RB2). (iii) Alternative nulls: $p_{\text{single}} = 1.00\%$ (symmetric triangular on $[0, W]$), 0.75% (Beta(2,2) scaled), 0.20% (log-uniform on $[1, W]$) — the uniform null is the declared one; the decision survives all three (RB3). None of these adjudicates the corner; the verdict rests on the pre-registered primary.

Appendix C — Pre-registration Certificate

This appendix is the chronology evidence package: what was frozen, in which document version, in which order, with the execution identifiers. It converts the §4 verdict from a declaration into a checkable record.

C.1 Target freeze (ZS-A31 v1.2 $\rightarrow$ v1.5, §6.3)

Exact target: $M_K/\bar{M}_P = e^{(-2\pi Q)}$, the 2π forced by Borchers–Wiesbrock, registered HYPOTHESIS-weak ($\approx 30\%$) in ZS-A31 as the single pre-registered candidate on the vacuum-energy line — with the explicit A31 discipline that the residual ($\Leftrightarrow C_{UV} \approx 1.25$ at 5.4% quartic gap) must not be absorbed into Q . At the time of this freeze *no formula universe existed*: the target predates U by construction.

C.2 Universe freeze (ZS-F38 v1.0 Appendix C, reconciled in ZS-F38 v1.1)

U of $|U| = 88$ formulas (the π -family 40, the A-powers 40, the telomere family 8, degeneracies retained); tolerance band $\frac{1}{4} \ln 4 \equiv C_{UV} \in [1/4, 4]$; single target $(c, q) = (2, 11)$; decision rule $p \leq 5\%$; primary statistic p_{single} , secondary p_{universe} (hierarchy made explicit in F38 v1.1); execution *deferred* to ZS-A32 by guard MC3. At the time of this freeze no execution existed.

C.3 Execution (this paper)

Identifier	Value
Script	zs_a32_verify_v1_1.py (supersedes v1_0; adds MC1 membership, MC7, RB1–RB3, ET1, CERT; all v1.0 values preserved)
Script SHA-256	eb142f7fef7cab78bef68f3129454dc97e89892f9362f91054e388ca8ba0bff8
RNG seed	11 (numpy default rng; analytic p-values are seed-independent)
Execution date	July 5, 2026 (UTC)
Primary result	$p_{\text{single}} = 0.500\%$ (analytic) / 0.499% (10^6 draws) $\leq 5\%$ — PASS

Declaration order	primary statistic and null range declared in-script above the first p evaluation (MC0, DECLARATIVE-GUARD); the documentary order C.1 → C.2 → C.3 is the load-bearing chronology (CERT1)
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Reading. Each freeze is verifiable against the cited corpus version independently of this paper; the certificate's claim is only the order. A critic who accepts the three version records must accept that the target was fixed before the universe and the universe before the execution — which is the entire content of “pre-registered”.

Appendix D — Export Theorem (corpus-independent statement)

For the external reader: the following is pure arithmetic, independent of every Z-Spin construction. It shows what is and is not being claimed when four hierarchy exponents are quoted.

Export Theorem. Let M be a mass unit (the reduced Planck mass), and suppose *one* registered hierarchy depth D and *one* finite UV factor C are given, with $M_{\text{eff}} = C^{(1/4)} M e^{(-D)}$, a dimensionless vacuum factor κ (here $\kappa = \frac{1}{2}(1260/4807)\omega^2 = 0.669$), a vacuum fraction Ω , and an age factor $\alpha = \text{Hot}_U$. Then Friedmann arithmetic forces, exactly (verified symbolically, check ET1):

$$\begin{aligned} M_{\text{eff}} / M &= C^{(1/4)} e^{(-D)} \\ \rho_{\Lambda} / M^4 &= \kappa C e^{(-4D)} \\ H_0 / M &= (\kappa/3\Omega)^{(1/2)} C^{(1/2)} e^{(-2D)} \\ t_U \cdot M &= \alpha (3\Omega/\kappa)^{(1/2)} C^{(-1/2)} e^{(2D)} \end{aligned}$$

Thus the apparent exponents $\{D, 2D, 4D\}$ are *not separate fits*: given the first line, the other three are consequences. The Z-Spin-specific content is exactly two statements — that $D = 2\pi Q$ with $Q = 11$ the register depth (the pre-registered hypothesis adjudicated in §4), and that κ is the corpus-computed 0.669 (the ZS-M44/F33/A31 chain). An external reader may reject both and still use the theorem: any theory that fixes one depth and one UV factor prices the wave, vacuum, Hubble, and age hierarchies simultaneously, and should be tested on the single depth, not credited four times.

References

- [1] ZS-F38 v1.0, The Register Clock Identity (Z-Spin corpus, July 2026); ZS-F38 v1.1 (registry reconciliation, (H-eval) decomposition, p-statistic hierarchy; frozen design and executed values preserved).
- [2] ZS-A31 v1.5, The Effective One-Parameter Reduction of the Z-Spin Vacuum-Energy Line (Z-Spin corpus, July 2026).
- [3] ZS-M46 v1.5; ZS-M47 v2.0 (Z-Spin corpus).
- [4] ZS-F37 v1.3, The Action-Level Two-Leg Law (Z-Spin corpus).
- [5] ZS-M44 v1.6 (integrating ZS-M45); ZS-F33 v1.8; ZS-F36 v2.1 (Z-Spin corpus).
- [6] ZS-A30 v2.1 ($\Omega_{\Lambda} = 83/121$); ZS-A27 v2.3 / ZS-A28 v2.0 (scale-generation no-go); ZS-A22 v2.2; ZS-A17 v1.5 (Z-Spin corpus).
- [7] ZS-M3 §8 / ZS-F10 (the telomere hierarchy $t_n = \bar{t}_P e^{(n\pi/A)}$); ZS-F24 v2.0, Table A1; ZS-S4; ZS-S7 (Z-Spin corpus).
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Version History

v1.0 (July 2026): Initial public release. Executes the ZS-F38 v1.0 Appendix C pre-registered Monte Carlo (primary $p = 0.50\% \leq 5\%$; corner (W) promoted HYPOTHESIS-weak $\rightarrow$ HYPOTHESIS-strong); states Theorem A32.1 (Friedmann-Forced Squares: the vacuum, length, and time corners as arithmetic consequences of the single wave-corner hypothesis, exponents $8\pi Q$ and $4\pi Q$, C_{UV} cross-corner identity to 0.06%) and Theorem A32.2 (the telomere rational $n_{\text{Hubble}} = 4QA = 1540/437$, the Friedmann-dressing identity to 3×10^{-5} , the A31 3.57 convention resolution); executes the model-level C_{int} ($\varepsilon = 0$ exact on the F38.T2 chain; physical C_{int} OPEN); separates the determinant and modular exponent families ($35\pi/3$ non-identification of record); registers the ZS-M47 §12.3 operator rows INHERITED-OPEN $\rightarrow$ ZS-M48/ZS-F39; updates B3 to a two-gate debt $\{(H\text{-CLK}), C_{UV}\text{-1PI}\}$. Verification 29/29 PASS + 6/6 guards. Zero new fitted Z-Spin parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. (Consolidated from internal Z-Spin Collaboration research notes following ZS-F38 v1.0.)

v1.1 (July 2026): External-review integration; verdict and all executed numbers unchanged. Adds Appendix C (Pre-registration Certificate: A31 v1.2 target freeze $\rightarrow$ F38 App. C universe freeze $\rightarrow$ A32 execution, with script SHA-256, seed, date) and Appendix D (Export Theorem, corpus-independent, symbolically verified ET1); hardens the MC suite (non-tautological MC1 target-membership check, MC7 degeneracy-retention check, MC0 renamed DECLARATIVE-GUARD) and adds the RB robustness audits (dedup invariance exact; simplest-core stratum coverage 3.50%; alternative nulls 1.00%/0.75%/0.20% — all audit basis, not verdict basis); reconciles all design citations to “first registered in ZS-F38 v1.0 Appendix C, reconciled in ZS-F38 v1.1”; states explicitly that the input is one Λ CDM background package ($H_0, \Omega_{\Lambda}, \rho_{\Lambda}$ Friedmann-related — the C_{UV} cross-corner agreement is a closure identity, not an additional datum); sharpens the Abstract (four hierarchy corners, one registered depth; particle floor separated); registers the C_{UV} Target Specification ($1.244 \pm \delta$, no Q-absorption) as the second gate’s contract and the $\varepsilon_{C_{\text{int}}}(\text{phys})$ discharge criterion for ZS-F39/M48. Verification 34/34 PASS + 7/7 guards (zs_a32_verify_v1_1.py). Zero new hypotheses.